

Sankara Narayanan Ramaligame

Software & IoT Engineer

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📍 Puducherry, India

SUMMARY

Software and IoT-focused B.Tech student with hands-on experience in full-stack development (Flutter, Spring Boot, Web Technologies) and embedded systems. Experienced in building scalable microservices, cross-platform applications, and integrating complex cryptographic operations with decentralized networking. Built and deployed a production website for HexHive Solutions.

PROJECTS

DAISY: Serverless E2EE Communication Tool

Security & Protocol Design | Nostr Relays, AES-256-GCM, ECDH, Web Workers

- Invented the **Daisy Key Exchange (DKE)**, a custom blind role negotiation protocol utilizing a lexicographical tie-breaker handshake to establish anonymous peer-to-peer connections.
- Implemented Web Workers for isolated cryptographic operations, integrating AES-256-GCM encryption and ephemeral ECDH key exchange for Perfect Forward Secrecy (PFS).
- Hardened application against state-machine exploits and metadata leakage via traffic obfuscation and automated red-team testing.

HexHive Solutions — Startup Company Website

Startup & Web Dev | React, JavaScript, SEO, Responsive Design

- Built the complete production website for HexHive Solutions — a tech career accelerator startup.
- Designed and shipped a production-grade React app with SEO optimization, responsive design, and modern UI/UX.

ML Fingerprint Gender Identification

AI / Computer Vision | Python, Machine Learning, TensorFlow

- Developed a machine learning application to identify gender based on fingerprint scans, achieving 80% accuracy.
- Processed datasets and trained identification models targeting publication in IEEE.

IoT Motion-Detecting Alarm System

Microservices & Infrastructure | ESP8266, Flask, Python, Render

- Developed a custom motion-detecting hardware monitoring system utilizing an ESP8266 module.
- Programmed backend infrastructure using Flask and deployed on Render with continuous health monitoring.

TECHNICAL SKILLS

Programming & Scripting

C++, Python, Java, Dart, JavaScript

Cryptography & Protocols

AES-256-GCM, ECDH, Nostr, DKE

Backend & Web

Spring Boot, Flask, Flutter, React

EDUCATION

B.Tech in Information Technology

Sri Manakula Vinayagar Engineering College (SMVEC)

Expected 2029 | Puducherry, India

ACHIEVEMENTS

- **Foundation for Excellence (FFE):** 2026 Scholarship Candidate.
- **OA Techfest:** Participant in regional technology and AI seminars.
- **HexHive Solutions:** Built & deployed full production website.